

Prime-Balanced Signed Incidence Witnesses for a Literal Prime-Shell Mask Defect

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Abstract

We continue the literal masked prime-shell analysis of TPC-348. Rather than testing one mask-hit coordinate at a time, we assign equal total positive and negative mass to the ordered shell primes and sum all divisibility incidences. The resulting deterministic vector has an exact cross-prime Gram expansion. For a nonzero incidence vector b_I and the defect block D_I , the induced Euclidean norm obeys

$$\|D_I\|_{2 \rightarrow 2} \geq \frac{\|D_I b_I\|_2}{\|b_I\|_2}.$$

On the locked two-origin, three-count, four-anchor, two-exponent, four-law panel, all 192 signed vectors are nonzero and have positive response. The signed response is 0.39083565842–0.954375010719 of the defect spectral norm, exceeds the best mask-hit coordinate baseline on 136 rows, and reaches one half of that norm on 175 rows. The comparison is not uniform: the signed vector loses to the coordinate baseline on 56 rows. These are finite observations, not a growing arithmetic estimate; the source-uniform masked L^2 problem and the twin-prime endpoint remain open.

1 Question and scope

TPC-348 localized the divisibility-mask defect with an exact coordinate lower witness. The next minimal question is whether the shell itself supplies a natural signed combination of hit coordinates that exposes cross-prime structure. We study one rule fixed before the finite audit: an equal positive and negative split of the ordered shell, with a neutral middle prime when the shell length is odd.

The exact algebra below is finite-dimensional. The numerical table is a declared panel audit only. In particular, no statement here supplies a source-native arithmetic vector, a uniform-in- x estimate, a fixed power of x , a Route-B reassembly, or a proof of the twin-prime conjecture.

2 The literal masked object

Let $I = \{o, o+1, \dots, o+M-1\}$, let R_I be restriction, and let E_I be zero extension. For an active shell prime p , write

$$(P_p f)(n) = \mathbf{1}_{p \nmid n} f(n)$$

for the divisibility projection. With the same kernel normalization as the parent releases, the physical and unmasked blocks are

$$A_I = \sum_p \varepsilon_p R_I P_p K_p P_p E_I,$$

$$T_I = R_I \left(\sum_p \varepsilon_p K_p \right) E_I, \quad D_I = A_I - T_I.$$

The source signs ε_p belong to the frozen four-law protocol. The new test vector below has its own declared coefficients β_j ; conflating these two sign systems would change the object.

3 Prime balance and the exact Gram interface

Order the shell as $p_0 < p_1 < \dots < p_{r-1}$ and put $m = \lfloor r/2 \rfloor$. Define

$$\beta_j = \begin{cases} +1, & 0 \leq j < m, \\ 0, & m \leq j < r - m, \\ -1, & r - m \leq j < r. \end{cases}$$

Thus there are equally many positive and negative coefficients, and at most one neutral coefficient. For $t \in I$, let

$$h_{p_j, I}(t) = \mathbf{1}_{p_j | t}, \quad b_I = \sum_{j=0}^{r-1} \beta_j h_{p_j, I}.$$

At a multiply-divisible position, all active incidences are added; this is an incidence contrast rather than an exclusive owner-class assignment. Whenever $b_I \neq 0$, set $x_I = b_I / \|b_I\|_2$.

Proposition 1 (prime balance). *The coefficient vector satisfies $\sum_j \beta_j = 0$ and has equal positive and negative support.*

Proof. The first and last blocks each have m indices and are disjoint. The middle block has size $r - 2m \in \{0, 1\}$. Summing the coefficients proves the claim. $\square$

Proposition 2 (prime-incidence Gram identity). *For every finite matrix D_I ,*

$$\|D_I b_I\|_2^2 = \sum_{j,k} \beta_j \beta_k \langle D_I h_{p_j, I}, D_I h_{p_k, I} \rangle. \quad (1)$$

Proof. By definition $b_I = \sum_j \beta_j h_{p_j, I}$. Apply linearity of D_I and expand the Euclidean inner product. The sum is finite, so no convergence assumption is involved. $\square$

Theorem 1 (signed-incidence lower witness). *If $b_I \neq 0$, then*

$$\|D_I\|_{2 \rightarrow 2} \geq \frac{\|D_I b_I\|_2}{\|b_I\|_2}. \quad (2)$$

Proof. The vector $x_I = b_I / \|b_I\|_2$ is a unit vector. Substitution into the definition of the induced Euclidean norm [1] gives

$$\|D_I\|_{2 \rightarrow 2} = \sup_{\|x\|_2=1} \|D_I x\|_2 \geq \|D_I x_I\|_2,$$

which is (2). $\square$

Remark 1. *The theorem is exact finite linear algebra. It does not say that the Gram cross terms in (1) have a common sign or that the balanced vector is close to a leading eigenvector.*

4 Frozen audit protocol

The producer locks the TPC-348 code and certificate. We retain

$$o \in \{40097, 48097\}, \quad M \in \{256, 512, 1024\}, \quad Q \in \{24, 36, 54, 80\}, \quad s \in \{1, 2\},$$

with height $H = 66$ and the four source sign laws `all_plus`, `alternating_index`, `mod4_character`, and `half_split`. There are $2 \cdot 3 \cdot 4 \cdot 2 \cdot 4 = 192$ rows.

For each row we form D_I , the incidence vector b_I , its normalized response, and the TPC-348 coordinate baseline

$$C_I = \max_{t \in J_I} \|D_I e_t\|_2, \quad J_I = \{t \in I : \exists p \mid t\}.$$

The baseline is used only for a finite comparison. The independent checker reverses the shell accumulation order, recomputes all row metrics, and checks the exact multi-hit anchor. The stress suite mutates the coefficient rule, incidence support, census, range, anchor, and firewall.

Table 1: TPC-349 prime-balanced signed-incidence audit.

Quantity	Certified finite readout
Rows	192
Nonzero signed vectors	192/192
Positive signed responses	192/192
Signed support per row	28–150
Incidence Gram records	192
Maximum Gram replay discrepancy	$1.06581410364 \times 10^{-14}$
$\ D_I x_I\ /\ D_I\ $	0.39083565842–0.954375010719
$\ D_I x_I\ /\ T_I\ $	0.0125941959067–0.430061305156
$\ D_I x_I\ /C_I$	0.542800508699–2.04702542827
Baseline beaten	136/192
At least half the defect norm	175/192

5 Finite results

Table 1 reports the canonical certificate. The denominator $\|T_I\|$ is the unmasked ideal comparison, while $\|D_I\|$ is the defect spectral norm.

The signed vector is often stronger than the best single coordinate, but the comparison is not uniform: 56 rows do not beat C_I . This is why the claim ledger marks a universal balanced-gain statement as `REFUTED_SCOPED`. The response/defect ratio itself is a valid finite lower-witness ratio because of Theorem 2; its persistence on growing panels is a separate open question.

5.1 Exact multi-hit anchor

Take $I = \{1, \dots, 14\}$, $Q = 4$, exponent one, and all-plus source signs. The shell is $(5, 7)$ and $\beta = (1, -1)$. The exact incidence vector is

$$b_I = (0, 0, 0, 0, 1, 0, -1, 0, 0, 1, 0, 0, 0, -1), \quad \|b_I\|_2^2 = 4.$$

The exact squared norm of its defect image is

$$\|D_I b_I\|_2^2 = \frac{1580136191762341638715051100269721298390649257672312877072677225319}{4277374121662663268940652066711233824047030196831076541809000000}.$$

The canonical certificate also stores the exact image-vector digest. This anchor includes positions divisible by both shell primes in the interval and therefore tests the incidence sum rather than a one-hit shortcut.

6 Adversarial checks and claim boundary

The normal and optimized producer outputs agree byte-for-byte. The reverse shell replay agrees with every stored metric within its declared floating tolerance, and the stress mutations reject a wrong beta split, an owner-only incidence vector, a changed census, a false universal-gain flag, a changed range, and a changed parent lock. These checks establish package integrity; they are not additional arithmetic theorems.

The strongest exact statements are the zero-sum coefficient identity, the incidence Gram expansion, and the normalized induced-norm lower witness. The strongest finite observation is the 136/192 baseline improvement. The main obstruction is that this improvement is not universal even on the declared panel. We claim neither a source-uniform masked operator bound nor an arithmetic L^2 cancellation estimate. The Session-named Route-A/Route-B evaluator files are absent from this checkout; the local bridge is explicitly fail-closed.

7 Conclusion and next question

TPC-349 supplies a reusable interface

$$\text{ordered shell} \longrightarrow \text{zero-sum beta} \longrightarrow \text{incidence contrast} \longrightarrow \text{prime Gram} \longrightarrow \text{finite norm witness}.$$

It demonstrates substantial finite response while exposing the limit of a single balanced rule: no uniform gain follows. The next minimal question is whether the signed incidence Gram persists under fresh growing panels and whether its cross-prime terms can be bounded without discarding the literal masks.

References

- [1] John B. Conway. *A Course in Functional Analysis*. Springer, 2nd edition, 2000.